QOSSAIN AWAIS

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PROFESSIONAL SUMMARY

AI researcher and founder building environmental AI systems, geospatial intelligence, medical imaging, and multi-agent architectures, with a growing focus on foundational AI architectures and AGI applications for scientific discovery. Collaborate with University of Oxford's NDORMS on medical imaging AI and co-authored a Springer Nature publication, both within my first two years of a Computer Science degree at Beaconhouse National University.

EDUCATION

Beaconhouse National University — Lahore, Pakistan
Bachelor of Computer Science — 2024 – Present

Superior College Pattoki — Pattoki, Punjab
Intermediate in Computer Science — Completed

PROJECTS & RESEARCH

Multi-Agent Autonomous Trading System — Completed
AI Research Project

- Built a multi-agent AI architecture using **Python, PyTorch, Scikit Learn and LLM APIs** to analyze satellite imagery and real-time financial data streams for autonomous trading signal generation.
- Developed autonomous specialized agents for computer vision, time-series forecasting, and LLM-based macroeconomic analysis across 3 independent data streams.
- Engineered a decision-fusion orchestrator combining multi-agent outputs into adaptive, feedback-driven trading signals.

Virtual AI Doctor (MSK X-ray Co-pilot) — MVP Phase
Medical AI Research Project | Supervisor: Dr. Usman Nazir, NDORMS, University of Oxford

- Building an MVP for AI-assisted detection of normal vs. abnormal findings in upper-limb musculoskeletal X-rays (Stanford MURA dataset, DenseNet-style CNN).
- Designing the system as a research co-pilot with a RAG-based medical knowledge layer and clinician-review safety framing — not autonomous diagnosis.
- Researching a 3D virtual doctor interface (React/Three.js) for explainable, patient-facing result presentation..

Pakistan Flood Early Warning System — Completed
Geospatial AI & Satellite Data Platform

- Developed a real-time flood prediction dashboard integrating 4 satellite and climate data sources (NASA SMAP, ERA5, Sentinel-1 SAR, OpenWeather) covering 17 districts across Pakistan.
- Implemented 3 scientific soil moisture prediction approaches validated against 2022 Pakistan flood data (33M affected), achieving AUC=0.93 with the combined FloodWatch index.
- Engineered pixel-level SMI heatmaps via Google Earth Engine with climate stripe visualizations and 4–6 week advance flood season forecasting.

Flood Mapping and Impact Analysis: Southeast Asia — Completed
Geospatial AI Project

- Processed satellite imagery across 5 South Asian countries (Pakistan, India, Nepal, Bangladesh, Bhutan) to detect and map flood-affected zones.
- Integrated population density layers to quantify human exposure and estimate risk distribution across affected regions.

Carbon Footprint Assessment: Beaconhouse National University Completed

Environmental Data Analysis Project

- Calculated institutional carbon emissions for BNU across 3 operational categories: energy consumption, transport, and facility management.
- Delivered a quantified emissions inventory supporting campus sustainability planning; findings contributed to a Springer Nature manuscript.

Hybrid Methane Plume Detection System Completed

Deep Learning & Physics-Based Remote Sensing Research

- Built a hybrid detection pipeline combining custom-trained deep learning models (ResNet34 classifier, F1=0.766; physics-informed UNet segmenter, IoU=0.543) with pretrained UNet++ networks (Sentinel-2, Landsat 8/9, EMIT) for global methane plume screening.
- Designed a physics-based multi-sensor verification engine in Google Earth Engine, fusing TROPOMI, ERA5 wind data, and Sentinel-2/Landsat SWIR anomalies (Beer-Lambert ΔXCH_4) to validate detections and eliminate false positives.
- Validated the system on confirmed super-emitter sites (Korpezhe, Aliso Canyon) with zero false positives across 10 negative-control sites and a year-long temporal scan; cross-validated against Carbon Mapper Tanager-1 hyperspectral data.
- Built an interactive React-Leaflet dashboard displaying real Sentinel-2 imagery, plume overlays, and per-model confidence breakdowns; deployed backend (FastAPI) and frontend via Docker on DigitalOcean.
- Submitted a paper based on this work as an author to the Visualising Climate conference.

Independent Research & Entrepreneurship

OpenAthena Founder, Data Intelligence Software MVP Phase

AI Research and Startup Project

- Building an autonomous intelligence platform for high stakes decision systems shipped an MVP covering Fraud, Risk, AML, and Internal/External Threat Detection.

Independent AI Architecture Research Research Phase

Self Research

- Studying new transformer/AI backbone architectures, with current focus areas in AGI applications for scientific discovery and grid optimization for sustainability.

PUBLICATIONS

Plant-Based Indoor Air Purification in the Pakistani Context Published

Discover Atmosphere, Springer Nature | Co-Author

- Contributed environmental impact analysis and carbon footprint evaluation to the manuscript.

Paper Link

Multi-Sensor Methane Plume Detection from Space Submitted

Visualising Climate Conference | Author

- Hybrid deep learning and physics-based methane plume detection system with interactive satellite dashboard.

EXPERIENCE

University of Oxford: NDORMS Remote / Oxford, UK

Research Collaborator | Supervisor: Dr. Usman Nazir (Data Scientist) 2025 – Present

- Collaborated with Dr. Usman Nazir (NDORMS, University of Oxford) on applied data science, environmental analysis and AI research.
- Contributed to data analysis pipelines, model development, and result interpretation under international

academic supervision.
- Gained direct exposure to Oxford-level research standards, reproducibility practices, and interdisciplinary AI applications in an international setting.

Beaconhouse National University Lahore, Pakistan
Research Assistant – Center for AI Research, BNU Summer 2025

- Built carbon calculation data pipelines to process energy consumption and operational records, producing a quantified carbon emissions data across 3 scopes.
- Analysed institutional carbon footprint of BNU using energy consumption, facility operations, and activity datasets; delivered actionable sustainability insights.
- Contributed findings directly to a peer-reviewed manuscript submitted to Discover Atmosphere, Springer Nature.

Knowledge Stream Lahore, Pakistan
Teaching Assistant — Advanced Excel & Power BI | Bank of Punjab Nov 2025 – Feb 2026

- Delivered hands-on instruction in Advanced Excel and Power BI to 40+ senior banking professionals at Knowledge Stream, a Bank of Punjab training program.
- Designed practical lab exercises covering data cleaning, pivot analysis, DAX formulas, and Power BI dashboard reporting.
- Improved student assessment outcomes by providing targeted one-on-one support and developing reusable training materials.

SKILLS

Programming: Python, React, HTML/CSS, JavaScript, TypeScript, Flask, Django.

AI / ML: Machine Learning, Deep Learning, AI System Design, Computer Vision, Remote Sensing, Multi-Agent Systems, LLMs, RAG.

Data & Analytics: Data Analysis, Data Visualization, Satellite Data Processing, GIS & Remote Sensing, Big Data.

Tools: VS Code, Antigravity, Git, GitHub, Google Earth Engine, QGIS, Power BI, Advanced Excel, Hugging Face, Ollama, Mistral, GCP, AWS, Pytorch, Scikit Learn, TensorFlow, Pandas, Numpy.
Languages: Urdu (Native), English (Professional)

ACHIEVEMENTS

- **1st Place**, Halshanas Hackathon — Google Developer Group (GDG) BNU, for the Hybrid Methane Plume Detection System.
- Selected as a research collaborator with the University of Oxford (NDORMS) in the first year of undergraduate study — an opportunity typically reserved for graduate researchers.
- Co-authored a Springer Nature manuscript and submitted a second paper to the Visualising Climate conference, both within the two years of my Computer Science degree.

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